MIL-O-18521(NOrd)

20 May 1955
SUPERSEDING
NAVORD OS 3013 A
29 JUNE 1945

MILITARY SPECIFICATION

OGIVE, PLASTIC, FOR FUZE MK 29

1. SCOPE

1.1 This specification governs the manufacture, assembly, and the preparation for delivery of Plastic Ogive for Fuze Mk 29, and the methods of inspection and tests upon which acceptance will be based. The ogive is used in the construction of Fuze Mk 29.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L-P-406	-- Plastics, Organic: General Specifications, Test Methods.
NN-B-621	-- Boxes, Wood, Nailed and Lock-Corner.
UU-C-843	-- Cushioning, Cellulosic.
LLL-B-631	-- Boxes; Fiber Corrugated (For Domestic Shipment).
PPP-B-601	-- Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-P-14	-- Plastic Materials, Molding and Plastic Parts, Molded; Thermosetting.
JAN-P-105	-- Packaging and Packing for Overseas Shipment-Boxes, Wood, Cleated, Plywood.

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JAN-P-108 -- Packaging and Packing for Overseas Shipment-Boxes, Fiberboard (V-board and W-board), Exterior and Interior.
JAN-P-133 -- Packaging and Packing for Overseas Shipment-Boxes, Set-Up, Paperboard.

STANDARDS

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

328558 -- Point Detonating Fuze Mk 29, and all documents listed thereon.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.

3.1.1 The plastic ogive covered by this specification shall be manufactured in accordance with Bureau of Ordnance Drawing 328558 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Material.--The plastic material used in molding the ogive shall conform to the requirements of Specification MIL-P-14, Type CFI-30 and in addition shall possess

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the following physical and chemical requirements when tested as specified in 4.5.1 through 4.5.4.

Specific Gravity	1.34 - 1.40
Inorganic Residue	0.0 - 2.0 percent
Acetone Extraction	1.30 - 2.50 percent
Moisture Absorption	0.30 - 1.20 percent

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", if a collection of units of the product used as a basis for government inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of 25 plastic ogives shall be delivered by the contractor for test to an agency designated by the Procuring Agency. These plastic ogives shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

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4.3 Production lot. -- For procurement purposes each lot of plastic ogives shall consist of 5000 units plus those required for all tests. Unless otherwise stated in the contract, order or requisition, all units for test purposes shall be at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes as defined in MIL-STD-105.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", Op 400; Government Inspection shall be in accordance with the provisions of general procedure for Government Inspection and Contractor's Inspection shall be in accordance with inspection by contractor, as contained in OP 400.

4.5 Test Procedures.

4.5.1 Specific gravity test. Two ogives chosen at random from each of 5000 ogives (maximum) shall be tested for specific gravity as specified in Federal Specification L-P-406, Method 5011. The ogives shall meet the specific gravity requirements of 3.2 for lot acceptance.

4.5.2 Inorganic residue test. -- Two ogives chosen at random from each lot of 5000 ogives (maximum) shall be tested for inorganic residue. The residue from the acetone extraction test of 4.5.3 shall be used for this test. The samples shall be carefully weighed separately after drying for 24 hours in an oven at 100 degrees C. They shall then be completely ashed in a muffle furnace at 750 degrees C, and the residue of each weighed separately. The inorganic residue shall be expressed as a percentage by weight, and shall be taken as the ratio of the weight of the residue to the weight of the specimen after drying at 100 degrees C for 24 hours. The ogives shall meet the inorganic residue requirements of 3.2 for lot acceptance.

4.5.3 Acetone extraction test. -- Two ogives chosen at random from each lot of 5000 ogives (maximum) shall be subjected to the acetone extraction test specified in Federal Specification L-P-406, Method 7021. The ogives shall meet the acetone extraction requirement of 3.2 for lot acceptance.

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4.5.4 Moisture absorption test. -- Two ogives chosen at random from each lot of 5000 ogives (maximum) shall be tested for moisture absorption. Weigh each ogive separately and place the samples in a temperature-humidity oven maintained at 25 degrees C and 70 percent relative humidity for 96 hours. The samples shall then be reweighed. The moisture absorption shall be expressed as a percentage by weight and shall be taken as the ratio of the gain in weight of the treated sample to the weight of the untreated sample. The ogives shall meet the moisture absorption requirements of 3.2 for lot acceptance.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Each ogive shall be wrapped in cellulosic cushioning conforming to the requirements of Federal Specification UU-C-843, and placed in a set-up paperboard box conforming to the requirements of Military Specification JAN-P-133.

5.1.2 Forty-eight boxes specified in 5.1.1 shall be packed in W5C fiberboard boxes conforming to the requirements of Military Specification JAN-P-108.

5.2 Packing. -- Four (4) boxes as specified in 5.1.2 shall be packed in a nailed wood or cleated-plywood box conforming to the requirements of Federal Specification PPP-B-601 or NN-B-621, respectively.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in compliance with Military Standard MIL-STD-129.

6. NOTES. -- There are none applicable.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government

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thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD. FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-4878

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